

Panel Data

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Introduction

Pooled Data

Fertility Rate

Wage

Incinerator

Fixed Effects
Model

Random
Effects Model

Panel Data

Jerome Dumortier

Required R packages

- [plm](#)
- [lmtest](#)

Documentation

- [Panel data econometrics in R](#)

Note regarding presentation of results

- Use of the package [stargazer](#) due to the large number of variables

Pooled vs. Panel Data

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Pooled data: Combination of multiple cross-sectional data over time

- Two or more different observational units over time
- Examples: Grades in an economics class from multiple semesters, [American Community Survey \(ACS\)](#), or General Social Survey (GSS)

Panel data: Repeated measurement of the same individual i over time t

- Individual units can be people, states, firms, counties, countries, etc.
- Necessary adjustments of standard error due to correlation across time

Assumptions about panel data and models

- Regular time intervals
- Errors are correlated
- Parameters may vary across individuals or time
- Intercept: Individual specific effects (fixed or random)

National Longitudinal Survey

- For NLSY79: Accessing data $\Rightarrow$ investigator $\Rightarrow$ Begin searching as guest $\Rightarrow$ Pick income as an example

Panel Study of Income Dynamics (PSID)

- Data on approximately 5,000 families on various socioeconomic and demographic variables

Survey of Income and Program Participation (SIPP)

- Interviews about economic condition of respondents

GSS is not a panel data set because different respondents are questioned every year

Advantages of Panel Data

Controlling for unobserved heterogeneity among observational units (e.g., firms)

- Time-invariant individual-specific effects (e.g., institutional, personality traits) that would otherwise bias estimates if omitted

Dynamic behavior

- Better understanding of the dynamic changes for observational units over time
- Combination of cross-sectional with time series data leading to more complete behavioral model

Efficiency of parameter estimates

- Increases in data variability improving estimates and avoiding multicollinearity

Less omitted variable bias

- Implicit inclusion of time-invariant, unobserved variables leading to less bias

Disadvantages of Panel Data

Data availability and collection complexity

- Very difficult and expensive to collect with potentially high attrition rates
- Accumulation of measurement error over time leading to estimation biases
- Example: NLSY79 requires tracking of individuals since 1979

Advanced estimation and interpretation

- High likelihood of autocorrelated errors and heteroskedastic
- Time-invariant variables are excluded in fixed effects models

Terminology and Types

Balanced versus unbalance panel

- A balanced panel has the same number of time-series observations for each subject or observational unit, whereas an unbalanced panel does not

Short versus long panel

- Short panel: Larger number of subjects or observational units than time periods.
- Long panel: Greater number of time periods than observational units

Types of regression models

- Pooled Ordinary Least Square model
- Fixed effects model
- Random effects model

Data `ferti11` from the GSS (1974–1984)

- *year*: 72 to 84, even
- *educ*: years of schooling
- *meduc* and *feduc*: mother's and father's education
- *kids*: number children ever born
- *east*, *northcentral*, and *west*: 1 if lived in at 16
- *farm*: 1 if on farm at 16
- *otherrural*: 1 if other rural at 16
- *town*: 1 if lived in town at 16
- *smallcity*: 1 if in small city at 16

Source: Jeffrey Wooldridge, Introductory Econometrics: A Modern Approach

Estimation Results

```
##
## =====
##                               Dependent variable:
##                               -----
##                               kids
## -----
## educ                -0.130*** (0.019)
## age                 0.499*** (0.141)
## I(age2)             -0.005*** (0.002)
## east                0.061 (0.133)
## northcentral        0.220* (0.121)
## west                0.051 (0.168)
## y82                 -0.414** (0.174)
## y84                 -0.565*** (0.177)
## Constant            -6.785** (3.099)
## -----
## Observations        1,129
## R2                  0.099
## Adjusted R2         0.086
## Residual Std. Error 1.581 (df = 1112)
## F Statistic         7.671*** (df = 16; 1112)
## =====
## Note:                *p<0.1; **p<0.05; ***p<0.01
```

Evolution of fertility rates over time after controlling of other observable factors:

- Base year: 1972
- Negative coefficients indicate a drop in fertility in the early 1980's
- Coefficient of $y82$ (-0.41) indicates that women had on average 0.41 less children, i.e., 100 women had 41 kids less than 1972
- This drop is independent from education since we are controlling for education.
- More educated women have fewer children
- Assumes that the effect of each explanatory variable remains constant.

Data cps7885 from the Current Population Survey:

- *educ*, *exper*, *union*, *female*, and year dummy variable *y85*

Interact year dummy with key explanatory variables to see if the effect of that variable has changed over time:

$$\ln(\text{wage}) = \beta_0 + \gamma_0 \cdot y85 + \beta_1 \cdot \text{educ} + \gamma_1 \cdot y85 \cdot \text{educ} + \beta_2 \cdot \text{exper} + \beta_3 \cdot \text{exper}^2 + \beta_4 \cdot \text{union} + \beta_5 \cdot \text{female} + \gamma_5 \cdot y85 \cdot \text{female}$$

```
cps7885$y85      = ifelse(cps7885$year==85,1,0)
bhat             = lm(log(wage)~y85+educ+y85:educ+exper+
                      I(exper^2)+union+female+y85:female,
                      data=cps7885)
```


Return to education

- β_1 is the return in 1978: 7.5%
- $\beta_1 + \gamma_1$ is the return in 1985: $7.5\% + 1.8\% = 9.3\%$

Change in the return of education

- γ_1 as the change in return over the 7 year period

Gender gap

- 1978 gender gap: 31.67%
- 1985 gender gap: $31.67\% - 8.51\% = 23.16\%$

Intercept

- β_0 and $\beta_0 + \gamma_0$ as the 1978 and 1985 intercepts, respectively

Data and Variables

Data set `kiel` about home values near the location of an garbage incinerator

- Run 1981 data
- Run 1978 data

Naive Implementation in R

```
kiel81      = subset(kiel, year==1981)
bhat81      = lm(rprice~nearinc, data=kiel81)
kiel78      = subset(kiel, year==1978)
bhat78      = lm(rprice~nearinc, data=kiel78)
```


To determine statistical significance

$$price = \beta_0 + \gamma_0 \cdot y81 + \beta_1 \cdot nearinc + \gamma_1 \cdot y81 \cdot nearinc$$

Interpretation

- β_0 : Average home value which is not near the garbage incinerator
- $\gamma_0 \cdot y81$: Average change in housing values for all homes
- $\beta_1 \cdot nearinc$: Location effect that is not due to the incinerator
- γ_1 : Decline in housing values due to incinerator

Implementation in R

```
bhat1      = lm(rprice~y81+nearinc+y81:nearinc,data=kiel)
bhat2      = lm(rprice~y81+nearinc+y81:nearinc+age+I(age^2)
               ,data=kiel)
bhat3      = lm(rprice~y81+nearinc+y81:nearinc+age+I(age^2)+
               cbd+rooms+area+land+baths,data=kiel)
```

[illegible]

Difference-in-difference estimator: $-\$30,688 - (-\$18,824) = -\$11,864$

$$\hat{\delta}_1 = (price_{81,near} - price_{81,far}) - (price_{78,near} - price_{78,far})$$

where $\hat{\delta}_1$ represents the difference over time in average differences in housing prices in the two locations.

The data set is used in many textbooks and comes also with the package [plm](#). Data on 10 companies over the period 1935 to 1954:

- *inv*: Investment
- *value*: Value of the firm
- *capital*: Capital stock

Companies of interest for this class: GM (firm 1), U.S. Steel (firm 2), GE (firm 3), Westinghouse (firm 8)

Pooling all cross-sectional and time series observations into a single data set and running an OLS regression.

$$inv_i = \beta_0 + \beta_1 \cdot value_i + \beta_2 \cdot capital_i$$

General formulation of the pooled model

$$y_{it} = \beta_0 + \beta_1 \cdot x_i + \epsilon_i$$

Issues with pooled OLS model:

- Ignores heterogeneity among the observations and time.

With heterogeneity: Biased and inconsistent estimates due to correlation between independent variables and error term.

Data Preparation and Pooled OLS Model

Use of firms 1, 2, 3, and 8:

```
grunfeld = subset(grunfeld,grunfeld$firm %in% c(1,2,3,8))
```

To use the functions from `plm`, define data as a panel data set:

```
grunfeld = pdata.frame(grunfeld,index=c("firm","year"))
```

Running a simple OLS model on the data:

- Using the regular `lm()` function
- Using the `plm()` function and specifying the model as **pooling**
- Name the outputs `grunfeld.ols` and `grunfeld.pooling`

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```
grunfeld.ols      = lm(inv~value+capital,data=grunfeld)
grunfeld.pooling  = plm(inv~value+capital,data=grunfeld,
                        model="pooling")
```


Theoretical Concepts I

Fixed effects model or Least-Squares Dummy Variable (LSDV) regression

- Constant slope coefficients but varying intercept over i

Regression equation:

$$inv_{it} = \beta_{0i} + \beta_1 \cdot value_{it} + \beta_2 \cdot capital_{it}$$

with $i = 1, 2, 3, 4$ and $t = 1, 2, \dots, 20$. This model can also be written as

$$inv_{it} = \alpha_0 + \alpha_1 \cdot D_{1i} + \alpha_2 \cdot D_{2i} + \alpha_3 \cdot D_{3i} + \beta_1 \cdot value_{it} + \beta_2 \cdot capital_{it}$$

Individual specific effects:

$$y_{it} = \alpha_i + \beta_i \cdot x_{it} + \epsilon_{it}$$

α_i can be fixed or random

Theoretical Concepts II

Fixed effects model

- Intercept β_{0i} is firm specific.
- For an individual, this could be education and/or ability, possibly correlated with independent variables
- Intercept is time-invariant.
- Slope coefficients do not vary across individuals (firms) or time

Implementation in R

```
grunfeld.fixed = plm(inv~value+capital,data=grunfeld,  
                     model="within")
```

```
stargazer(grunfeld.fixed,no.space=TRUE,
          single.row=TRUE,type="text")
```

```
##
## =====
##                Dependent variable:
##                -----
##                inv
## -----
## value           0.108*** (0.018)
## capital         0.345*** (0.027)
## -----
## Observations           80
## R2                     0.806
## Adjusted R2            0.792
## F Statistic    153.291*** (df = 2; 74)
## =====
## Note:          *p<0.1; **p<0.05; ***p<0.01
```

Firm-Specific Intercepts and Hypothesis Test

In order to get the firm specific intercepts:

```
fixef(grunfeld.fixed)
```

```
##          1          2          3          8
## -85.515   94.988 -246.228  -59.386
```

Testing whether a fixed effects or OLS is appropriate (H_0 : OLS better):

```
pFtest(grunfeld.fixed,grunfeld.ols)
```

```
##
## F test for individual effects
##
## data:  inv ~ value + capital
## F = 67.215, df1 = 3, df2 = 74, p-value < 2.2e-16
## alternative hypothesis: significant effects
```

If the p-value is below 0.05 then the fixed effects model is a better choice.

Implementation in R using `lm()`

Implementation of a fixed effects model with the command `lm()`

```
bhat = lm(inv~value+capital+factor(firm),data=grunfeld)
```

The general fixed effects model can be expressed as

$$y_{it} = \beta_{0i} + \beta_1 \cdot x_{1,it} + \beta_2 \cdot x_{2,it} + \epsilon_{it}$$

Instead of treating β_{0i} as fixed, the random model assumes

$$\beta_{0i} = \beta_0 + v_i$$

where v_i is random error term with a mean of zero and variance σ_v^2 . According to Gujarati: *What we are essentially saying is that the four firms included in our sample are a drawing from a much larger universe of such companies and that they have a common mean value for the intercept β_0 and the individual differences in the intercept values of each company are reflected in the error term v_i .*

```
grunfeld.random = plm(inv~value+capital,data=grunfeld,model="random")
stargazer(grunfeld.random,no.space=TRUE,single.row=TRUE,type="text")
```

```
##
## =====
##                Dependent variable:
##                -----
##                inv
## -----
## value                0.108*** (0.017)
## capital              0.345*** (0.027)
## Constant             -73.085 (81.172)
## -----
## Observations                80
## R2                          0.803
## Adjusted R2                 0.798
## F Statistic                 314.851***
## =====
## Note:          *p<0.1; **p<0.05; ***p<0.01
```

Breusch-Pagan Lagrange Multiplier (LM) Test

For random effects models: Null hypothesis of no panel effect, i.e., OLS is better. If p-value is below 0.05, we reject the null hypothesis and thus, a random effects model is more appropriate than the OLS.

```
plmtest(grunfeld.pooling,type=c("bp"))
```

```
##  
##  Lagrange Multiplier Test - (Breusch-Pagan)  
##  
## data:  inv ~ value + capital  
## chisq = 378.44, df = 1, p-value < 2.2e-16  
## alternative hypothesis: significant effects
```

Hausman Test: Fixed or Random Model

The Hausman Test tests the null hypothesis that the preferred model is a random effects model. It basically tests whether the unique errors are correlated with the regressors.

```
phtest(grunfeld.random,grunfeld.fixed)
```

```
##
```

```
##  Hausman Test
```

```
##
```

```
## data:  inv ~ value + capital
```

```
## chisq = 0.14882, df = 2, p-value = 0.9283
```

```
## alternative hypothesis: one model is inconsistent
```

Testing for Heteroscedasticity

```
bptest(inv~value+capital+factor(firm),data=grunfeld)
```

```
##
```

```
##   studentized Breusch-Pagan test
```

```
##
```

```
## data:  inv ~ value + capital + factor(firm)
```

```
## BP = 25.375, df = 5, p-value = 0.0001179
```

If the p-value is below 0.05, then we have heteroscedasticity.

Heteroscedasticity Consistent Coefficients and Standard Errors

```
coeftest(grunfeld.fixed,vcovHC)
```

```
##
```

```
## t test of coefficients:
```

```
##
```

```
##          Estimate Std. Error t value  Pr(>|t|)
```

```
## value    0.108400   0.014293  7.5839 7.902e-11 ***
```

```
## capital  0.345058   0.031152 11.0765 < 2.2e-16 ***
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```